

because $\tilde{u} > \tilde{c}$. Thus, as $r \rightarrow 0$, ignoring is a better response than sharing. But note that $U_i(\mathcal{D}) = \tilde{u}(1 - \pi_i) - \tilde{c} \stackrel{r \rightarrow 0}{=} \tilde{u} - \tilde{c} > 0$, so as $r \rightarrow 0$, disliking is a better response than ignoring. As a result, disliking is a best response for any fixed $b < 1$ as $r \rightarrow 0$. The claim in (ii) thus follows from continuity of equation (2). ■

A.2 Proofs from Section 3

Proof of Theorem 1. Claim (ii) follows directly from Lemma A.1 and establishes that the Bayesian-Nash equilibria are exactly the fixed points of the map ψ . Clearly the cutoff space $\mathbf{B}$ is convex and compact (it is defined by $[0, 1]^{2N}$). To see that ψ is continuous, notice that for $\psi : (\mathbf{b}^*, \mathbf{b}^{**}) \mapsto (\mathbf{b}^{*,BR}, \mathbf{b}^{**,BR})$, $\mathbb{E}_{\mathbf{P}(\mathbf{b}^*, \mathbf{b}^{**})}[U_i^{(2)}]$ is continuous because H is continuous (and $U_i(\mathcal{D})$, $U_i(\mathcal{I})$, and $U_i^{(1)}$ do not depend on $(\mathbf{b}^*, \mathbf{b}^{**})$). Moreover, by the same reasoning as in Lemma A.2, $U_i^{\mathbf{P}(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S})$ and $U_i^{\mathbf{P}(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S}) - U_i(\mathcal{D})$ are monotone and continuous. Because these expressions are continuous in $(\mathbf{b}^*, \mathbf{b}^{**})$, the corresponding best-response cutoffs, $(\mathbf{b}^{*,BR}, \mathbf{b}^{**,BR})$ are also continuous in $(\mathbf{b}^*, \mathbf{b}^{**})$. By Brouwer's fixed-point theorem, there exists a Bayesian-Nash equilibrium, proving (i).

Finally, noting that the cutoff space $\mathbf{B}$ is a complete lattice and ψ preserves the component-wise order $\succeq$ (by Lemma A.2), Tarski's fixed-point theorem establishes that the set of equilibrium cutoffs forms a lattice (see Tarski (1955)). By definition of a lattice order, there exists a least-sharing equilibrium (largest $\mathbf{b}^{**}$) and a most-sharing equilibrium (smallest $\mathbf{b}^{**}$). ■

Proof of Proposition 1. Recall that π_i is given by equation (2) and provides the (ex-post) belief of the article's veracity conditional on observables (r, m) . Also observe that:

$$\frac{\partial \pi_i}{\partial r} = \frac{(1 - b_i + p(2b_i - 1))(1 - b_i + q(2b_i - 1))}{(1 - b_i + q(1 - \phi(r))(2b_i - 1) - p(\phi(r) - 2\phi(r)b_i))^2} \phi'(r).$$

Because $\phi'(r) > 0$, it is clear that when $b_i > 1/2$, $\partial \pi_i / \partial r > 0$. When $b_i < 1/2$, $1 - b_i + p(2b_i - 1)$ is minimized when $p = 1$, in which case it is equal to $b_i \geq 0$ (and with this inequality strict whenever $p < 1$). Similarly, when $b_i < 1/2$, $1 - b_i + q(2b_i - 1)$ is minimized when $q = 1/2$, in which case it is equal to $1/2 > 0$. Thus, $\partial \pi_i / \partial r > 0$ for all b_i .

Similarly, when $\phi' \geq \phi$, a given reliability score r with misinformation structure ϕ' can be translated into a higher reliability score $r' \geq r$ but with misinformation structure ϕ (because both ϕ, ϕ' are monotonically increasing). As a consequence, a decrease in misinformation is isomorphic to greater reliability of the articles. It is thus sufficient to prove the latter leads to uniformly more sharing in both the least and the most sharing equilibria.

Note that the social media game is supermodular and has increasing differences in reputability. To see this, note that for all $r' \geq r$:

$$[U_i(\mathcal{S}, r') - U_i(\mathcal{I}, r')] - [U_i(\mathcal{S}, r) - U_i(\mathcal{I}, r)] = U_i(\mathcal{S}, r') - U_i(\mathcal{S}, r) = U_i^{(1)}(r') - U_i^{(1)}(r) = (u + c)(\pi_i(r') - \pi_i(r)),$$